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(54) **DUAL OUTPUT UNINTERRUPTIBLE
POWER SUPPLY**

(56) **References Cited**

U.S. PATENT DOCUMENTS

(71) Applicant: **Microsoft Technology Licensing, LLC**,
Redmond, WA (US)

5,994,795 A * 11/1999 Gabillet H02J 9/062
307/26

(72) Inventors: **Banha Sok**, Kirkland, WA (US);
Rameez K. Kazi, Bellevue, WA (US);
Shaun L. Harris, Redmond, WA (US);
John J. Siegler, Redmond, WA (US)

9,680,333 B1 6/2017 Brooks et al.
2016/0285267 A1* 9/2016 Stone G06F 1/3203
2016/0372971 A1* 12/2016 Budde H02M 5/42

FOREIGN PATENT DOCUMENTS

(73) Assignee: **Microsoft Technology Licensing,
LLC.**, Redmond, WA (US)

CN 110999012 A 4/2020

OTHER PUBLICATIONS

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U.S.C. 154(b) by 0 days.

Office Action Received for Indian Application No. 202247018049,
mailed on Dec. 2, 2024, 6 Pages.

Office Action Received for Chinese Application No. 202080076266.
8, mailed on Mar. 4, 2025, 25 pages. (English Translation Provided).

(Continued)

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Primary Examiner — Lincoln D Donovan

Assistant Examiner — Alex W Lam

(74) *Attorney, Agent, or Firm* — Barta Jones, PLLC

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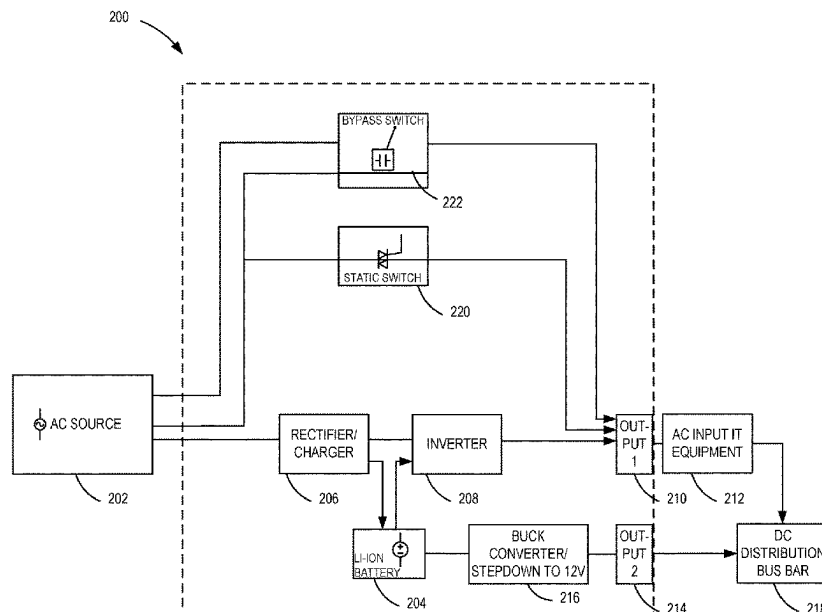
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H02J 9/061; G06F 1/30
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(57) **ABSTRACT**

An uninterruptible power supply (UPS) includes a rectifier configured to receive alternating current (AC) power. The UPS further includes a first output connected to the rectifier through an inverter. The first output is configured to output an AC power supply. The UPS also includes a second output connected to the rectifier through a battery backup and a stepdown converter. The second output is configured to output a direct current (DC) power supply in response to a detected power anomaly condition, thereby providing extra redundancy that allows for increased power availability and uptime.

20 Claims, 7 Drawing Sheets



(56)

References Cited

OTHER PUBLICATIONS

Communication pursuant to Rules 70(2) and 70a(2) EPC and reference to Rule 39(1) Received in European Patent Application No. 24216950.6, mailed on May 7, 2025, 2 pages.

Communication pursuant to Article 94(3) EPC, Received for European Application No. 20811156.7, mailed on Apr. 30, 2025, 4 pages.

* cited by examiner

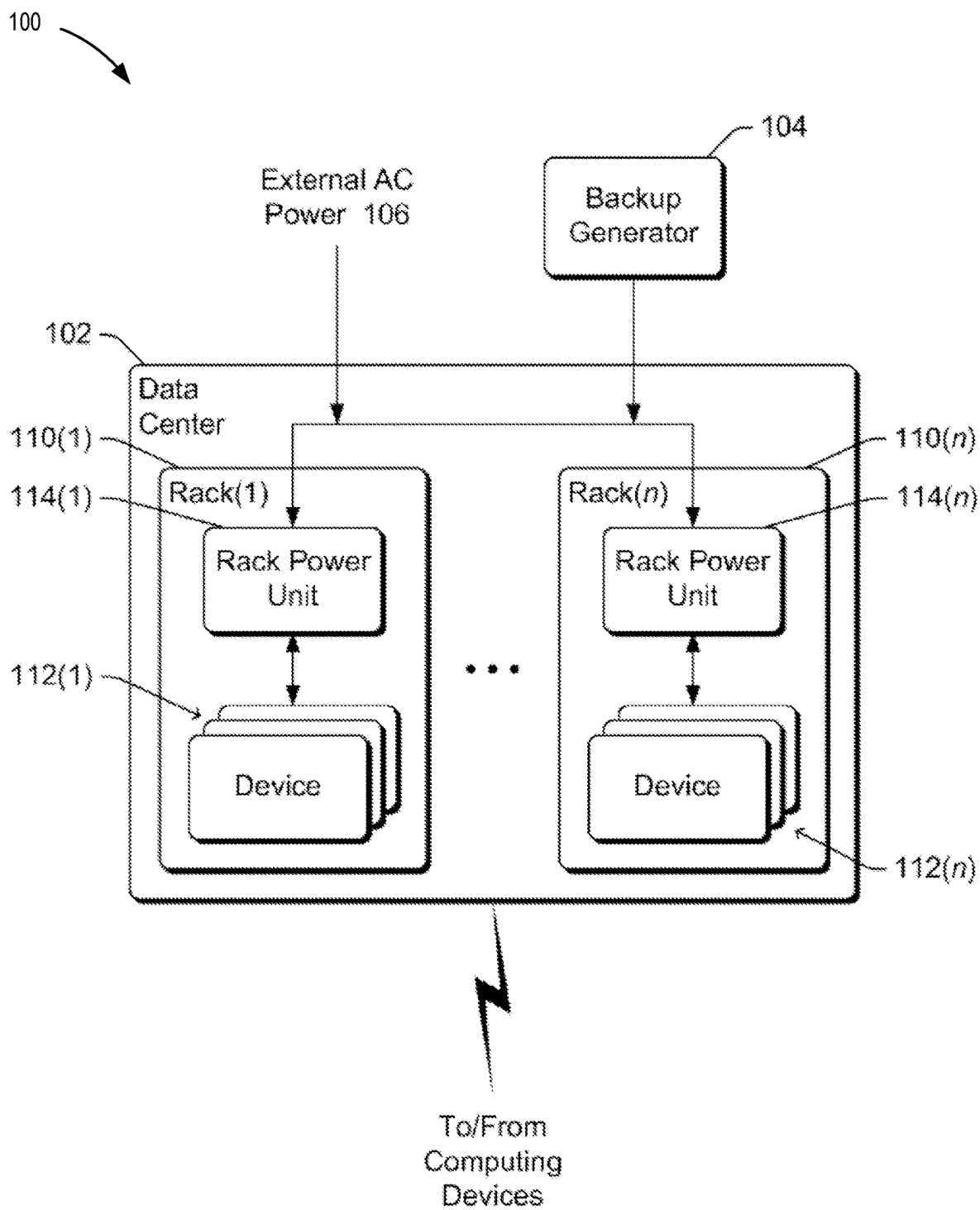


FIG. 1

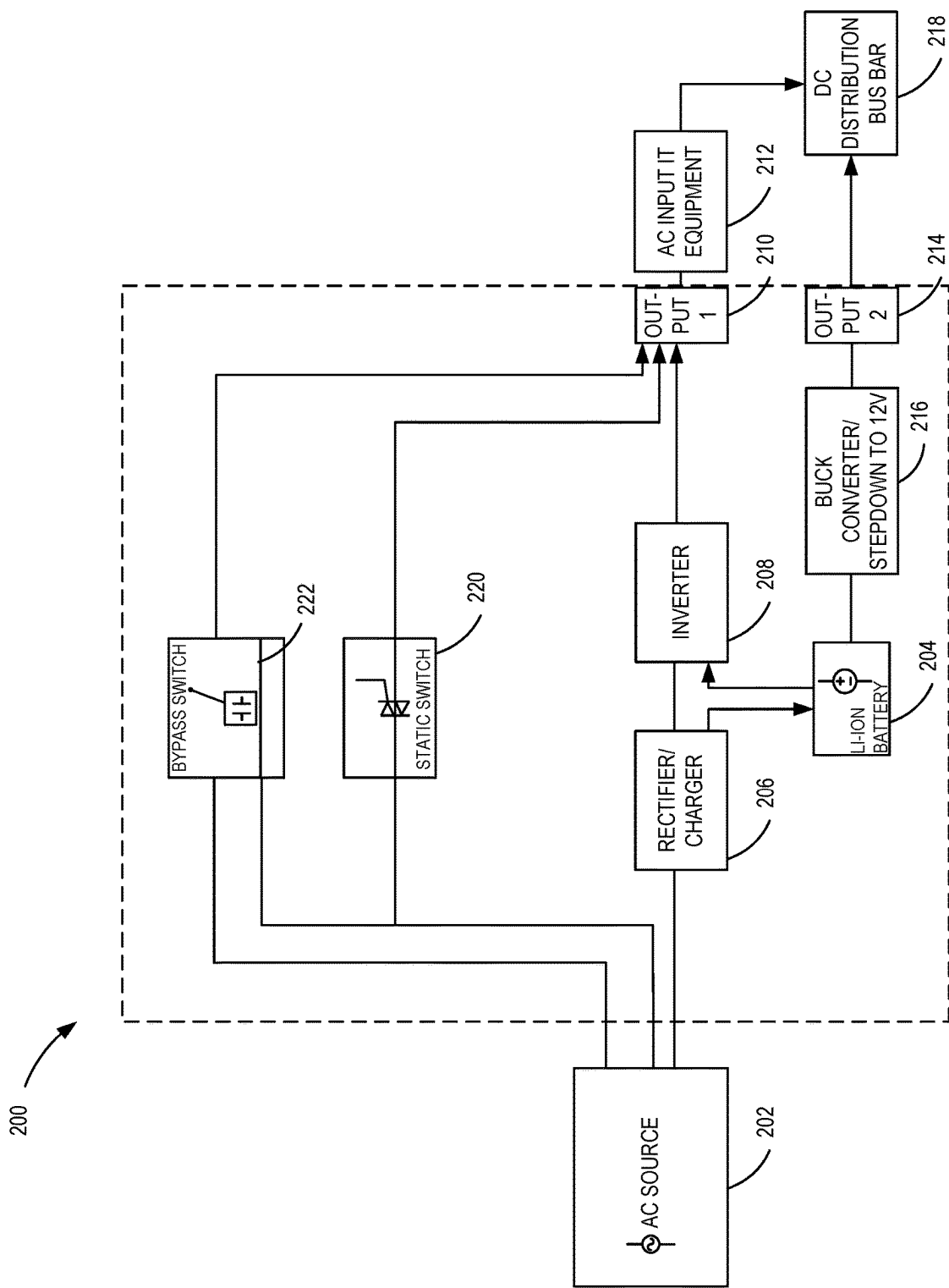


FIG. 2

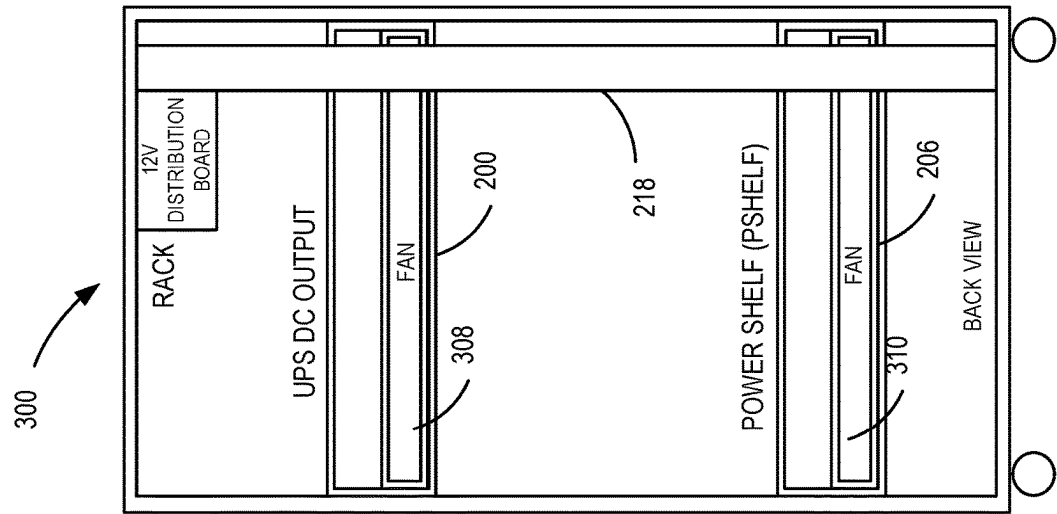


FIG. 4

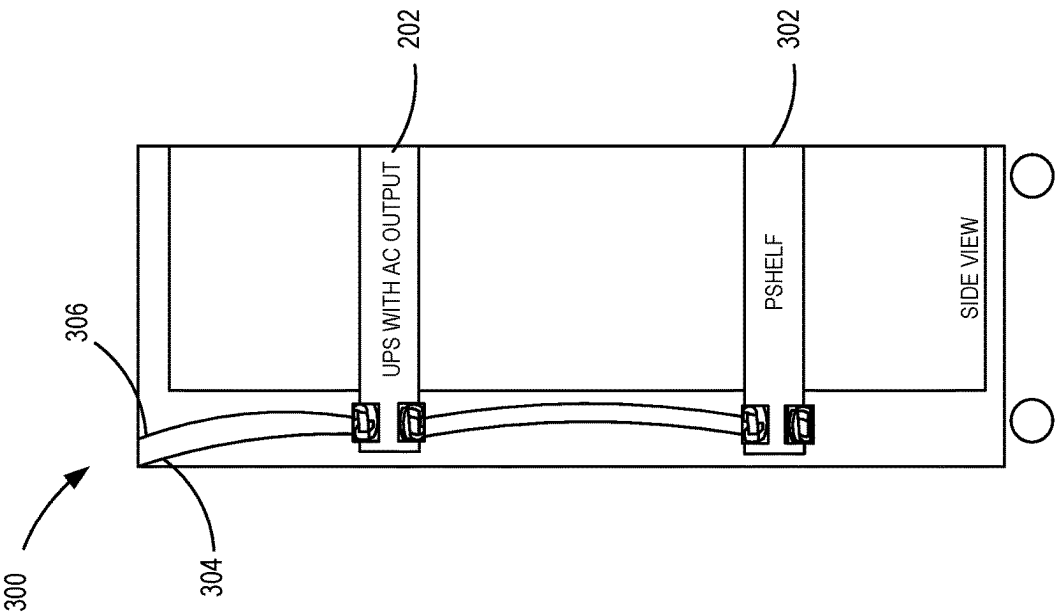


FIG. 3

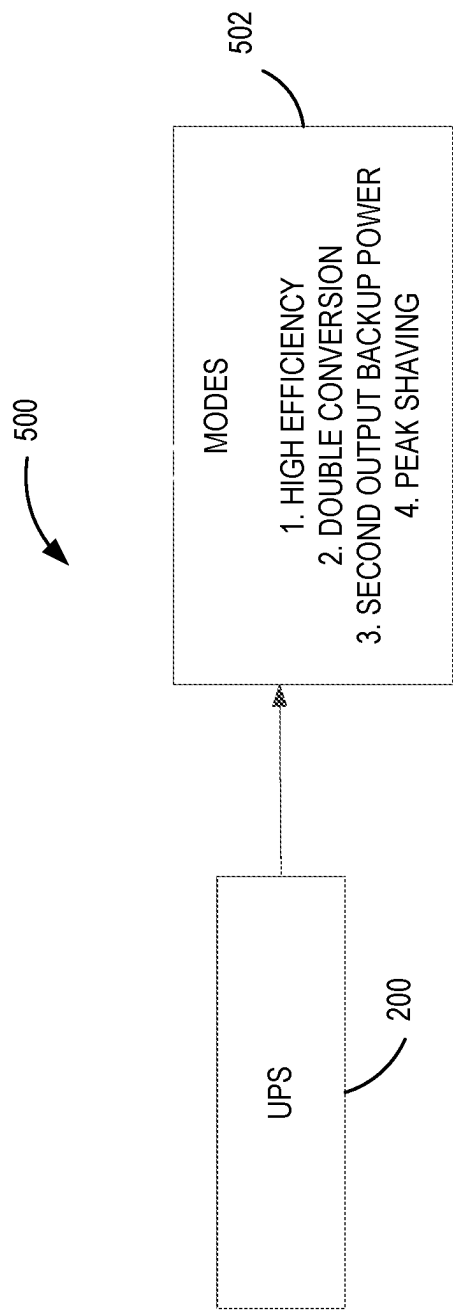


FIG. 5

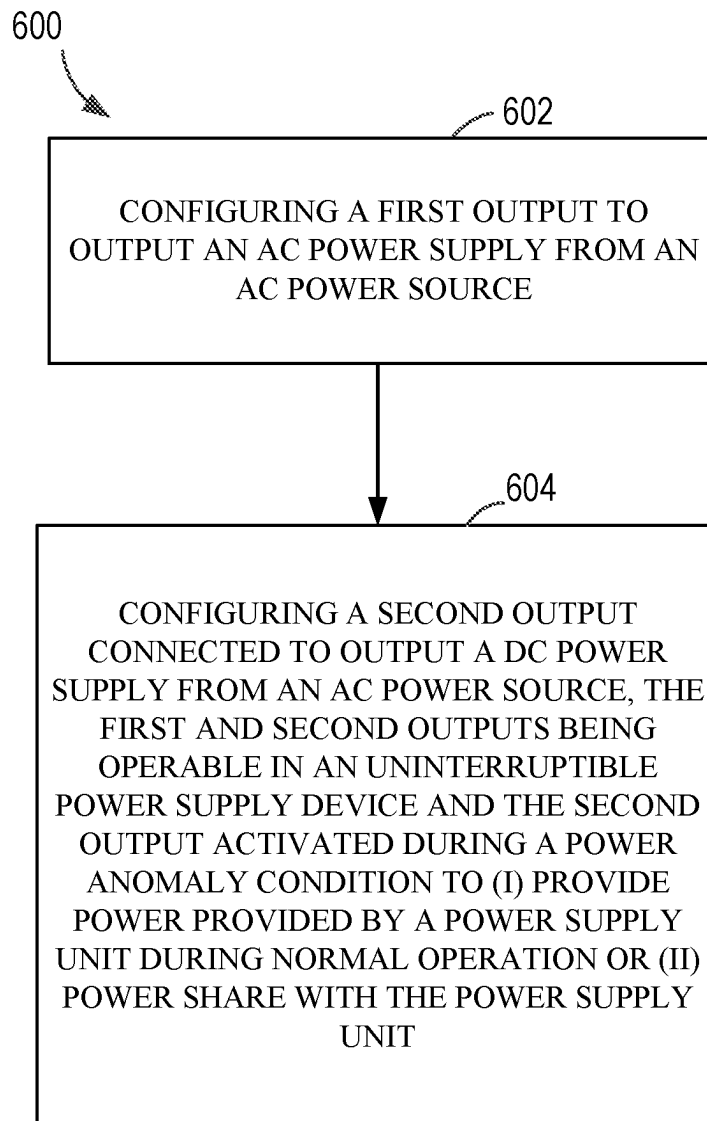


FIG. 6

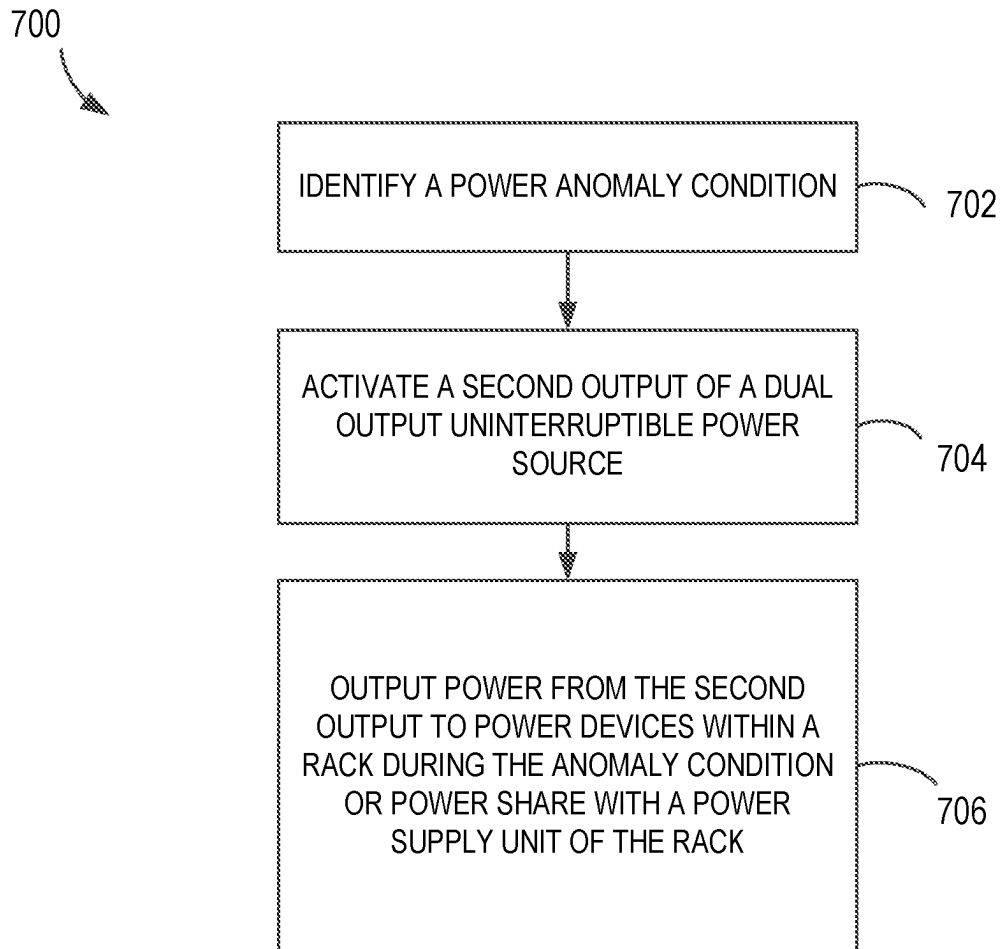


FIG. 7

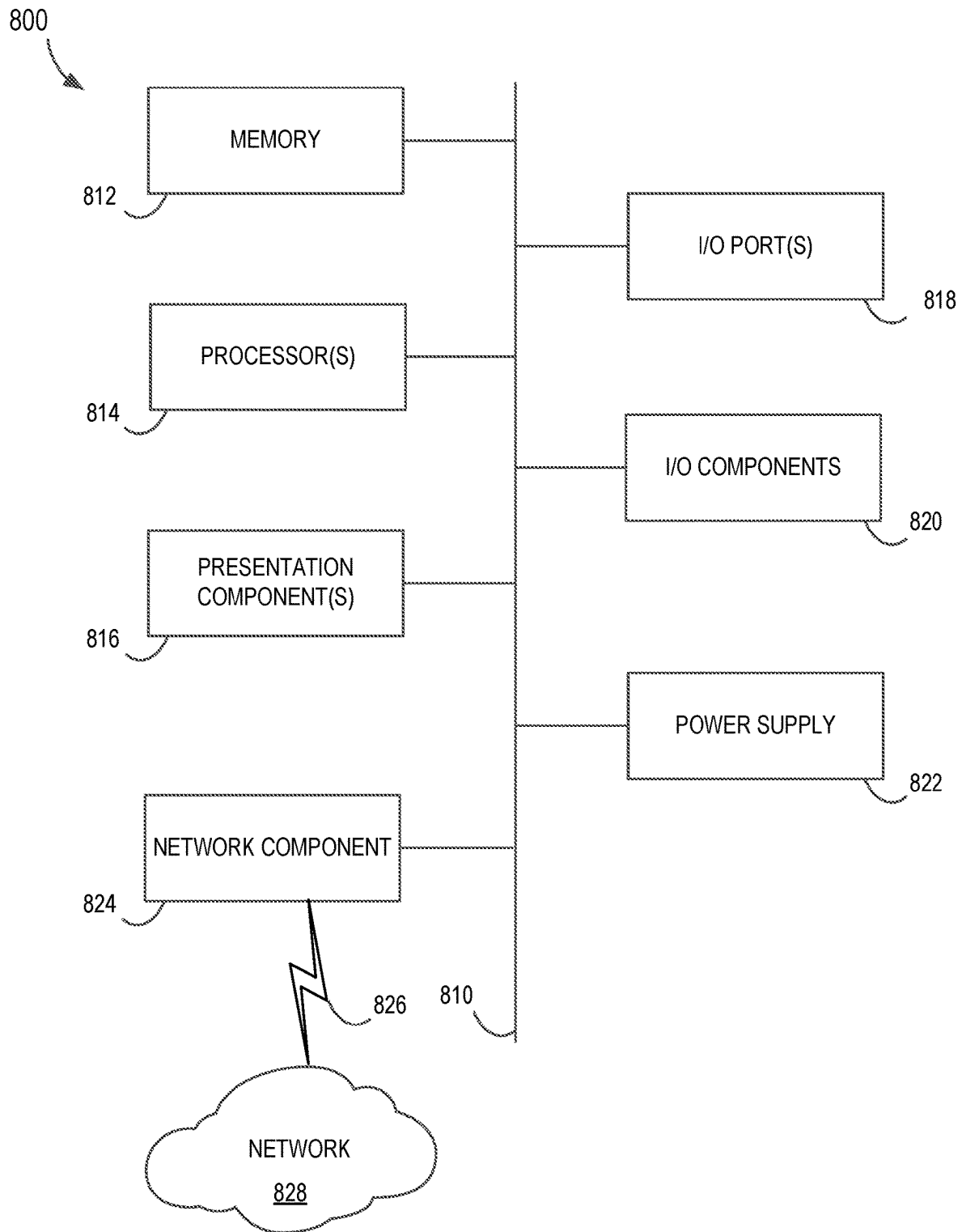


FIG. 8

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**DUAL OUTPUT UNINTERRUPTIBLE
POWER SUPPLY****CROSS-REFERENCE TO RELATED
APPLICATION**

This application is a continuation application of and claims priority to U.S. patent application Ser. No. 16/675, 213, entitled "DUAL OUTPUT UNINTERRUPTIBLE POWER SUPPLY," filed on Nov. 5, 2019, the disclosure of which is incorporated herein by reference in its entirety.

BACKGROUND

In certain environments, a number of computing devices operate together at a particular location to provide a service, such as data centers or server farms providing services over the Internet. It is desirable to have a constant source of power at such locations so that the computers and other devices operating at that location continue functioning. Providing a constant source of power, however, can be difficult because power providers occasionally experience power outages or other power failures occur. As such, a backup power supply to protect against such a power outage is beneficial.

Backup power can be provided by an uninterruptible power source or uninterruptible power supply (UPS). The UPS provides emergency power to a load (such as the computing devices) when the input power source or main power fails. The UPS provides near-instantaneous protection from input power interruptions, by supplying energy stored, for example, in batteries.

In a standard Information Technology (IT) equipment rack in a data center, AC power goes into the UPS, which outputs AC to a power distribution unit (PDU) to be distributed to all the power supply units (PSUs) in the rack. In recent years, power supply arrangements have been configured in a power shelf infrastructure. The power shelf is a rackmount system used in data center racks designed to provide power rectification, system management and power distribution. However, current power shelf architectures sometimes do provide power backup redundancy that ensures a long enough power supply during a power failure, which would be beneficial to ensure constant power (at least for a short time period), such as to critical IT equipment in the data center during the power failure.

SUMMARY

This Summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description. This Summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used as an aid in determining the scope of the claimed subject matter.

An uninterruptible power supply (UPS) comprises a rectifier configured to receive alternating current (AC) power. The UPS further comprises a first output connected to the rectifier through an inverter. The first output is configured to output an AC power supply. The UPS also comprises a second output connected to the rectifier through a battery backup and a stepdown converter. The second output is configured to output a direct current (DC) power supply in response to a detected power anomaly condition.

Many of the attendant features will be more readily appreciated as the same becomes better understood by

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reference to the following detailed description considered in connection with the accompanying drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

The present description will be better understood from the following detailed description read in light of the accompanying drawings, wherein:

FIG. 1 is a block diagram illustrating a system having a UPS according to an embodiment;

FIG. 2 is a schematic block diagram of a UPS according to an embodiment;

FIG. 3 is a block diagram illustrating a side view of an Internet Technology (IT) rack according to an embodiment;

FIG. 4 is a block diagram illustrating a back view of the IT rack of FIG. 3;

FIG. 5 illustrates multi-mode operation of a UPS according to an embodiment;

FIG. 6 is a flowchart of a process for providing a UPS according to an embodiment;

FIG. 7 is a flowchart of a process for providing power during power anomaly conditions according to an embodiment; and

FIG. 8 is a block diagram of an example computing environment suitable for implementing some of the various examples disclosed herein.

Corresponding reference characters indicate corresponding parts throughout the drawings. In the figures, the systems are illustrated as schematic drawings. The drawings may not be to scale.

DETAILED DESCRIPTION

The systems and methods described herein are configured to provide a dual output UPS that has extra redundancy. In one implementation, a multi-mode system UPS includes two outputs to accommodate different power outage conditions and can sustain a power outage for a time period (e.g., five minutes) to allow, for example, critical operations to be performed before system shutdown, switchover, etc. The UPS is operable in different modes, including a high efficiency mode and double conversion mode (input instability),

In some examples, an alternating current/direct current (AC/DC) dual output UPS is configured for operation in a power shelf rack. In one particular implementation, redundancy is provided such that the system does not rely solely on the power shelf to supply power (e.g., 12V power), but the UPS is also able to provide 12V power in the case of power anomalies (e.g., power outage to the power shelf). As a result, power availability/uptime is increased. Thus, an additional redundancy is thereby provided.

FIG. 1 illustrates an example system 100 implementing a rack-based UPS in accordance with one or more implementations. In one particular example, a multi-mode UPS includes a first output and a second output. The multi-mode UPS may be configured for operation in a data center, such as a 17 KW UPS operable in a high efficiency mode and double conversion mode (input instability), as well as configured to sustain a power outage for a period of time.

More particularly, and continuing with the data center example, the data center includes multiple computing devices and optionally other networking devices that are located within a device rack. AC power is provided to the data center from an external power source, and also from a backup generator in the event of a loss of power from the external power source. Additionally, each device rack has a rack power unit that receives the AC power and converts the

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AC power to DC power, which in turn is provided to the devices located within that device rack. Each rack power unit also includes one or more batteries or battery packs that provide power to the devices within that rack during a time period between the loss of power from the external AC power source and, for example, the backup generator becoming operational and providing AC power to the data center. Each rack power unit can also provide additional functionality. For example, the UPS is operable to output the AC output to the power shelf while switching off the second output. In the case of a power anomaly condition, the second output provides a source of DC power to a bus bar that can be current shared with the power shelf output or operate alone as discussed herein.

With particular reference to the configuration illustrated in FIG. 1, the system 100 includes a data center 102, a backup generator 104, and external AC power 106 received from an external power source. The data center 102 includes one or more (n) device racks 110(1), . . . , 110(n), each including one or more devices 112 and a rack power unit 114. The device racks 110 are also referred to herein as racks. It should be noted that although reference is made herein to device racks, the devices 112 and power units 114 can alternatively be grouped into other containers, mounting units, or other grouping configurations. In such configurations, the rack-based UPS techniques discussed herein can be implemented based on such alternate groupings rather than based on a rack.

The data center 102 operates to provide one or more services to various computing devices. The computing devices can be located in close physical proximity to the data center 102, and/or located across a wide geographic range (e.g., throughout a country or throughout the world). The data center 102 communicates with the computing devices via a variety of different networks, including the Internet, a local area network (LAN), a cellular or other phone network, an intranet, other public and/or proprietary networks, combinations thereof, and so forth. The data center 102 can be accessed by a variety of different types of computing devices, such as a desktop computer, a laptop computer, a mobile station, an entertainment appliance, a television, a set-top box communicatively coupled to a display device, a cellular or other wireless phone, a game console, an automotive computer, and so forth.

The data center 102 can provide one or more of a variety of different services to computing devices. In one example, the data center 102 provides one or more of a social networking service, an email service, a search service, an information resource/storage service, a messaging service, an image and/or video sharing service, a gaming or other entertainment service, and so forth. The one or more services provided by the data center 102 can be publicly available or alternatively access to one or more of the services can be restricted to particular users (e.g., those having a valid account as verified by a service of data center 102).

In the system 100, external AC power 106 is power received from one or more external power sources, such as a power station managed by a power utility company. The external AC power 106 is, for example, single-phase or 3-phase power. An interruption in external AC power 106 (also referred to as a power outage or power failure) can occur, and is a situation where the expected external AC power 106 is not received by the data center 102. A variety of causes exist for such an interruption, such as a failure at a power station that provides the power 106, a failure in a power transmission line between the power station and the

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data center 102, among others. In some examples, the interruption can occur as a result in a failure in a component of the system 100.

In the illustrated example, the backup generator 104 is a power generator that operates as a backup source of AC power in the event of an interruption in the external AC power 106. The backup generator 104 can be, for example, a diesel-powered or gas-powered generator. Although a single backup generator 104 is illustrated in the system 100, optionally multiple backup generators 104 (e.g., each responsible for providing AC power to one or more racks 110) can be included in system 100. The backup generator 104 can provide, for example, single-phase or 3-phase AC power, typically providing the same single-phase or 3-phase power as the external AC power 106. Alternatively, the backup generator 104 provides DC power rather than AC power.

The backup generator 104 in some examples is located in close physical proximity to the data center 102. In one example, a controller in the backup generator 104, or alternatively in another component or device, detects an interruption in the external AC power 106. In response to the detected interruption in the external AC power 106, the backup generator 104 is powered on and begins generating AC power to provide to the data center 102. Typically, there is a time period between the interruption in the external AC power 106 and the backup generator 104 generating sufficient AC power to power the data center 102 (at which point the backup generator 104 is referred to as being online). This time period can vary based on one or more of the manner in which the interruption of power is detected, the power used by the data center 102, and the particular backup generator 104. The rack power units 114 within the individual racks of data center 102 provide power to the devices 112 in the device racks 110 during this time period, as discussed in more detail below.

Multiple devices 112 in the data center 102 operate to provide the functionality of the one or more services provided by the data center 102. A variety of different types of devices can be included. The devices 112 include one or more server computers, such as rack servers or blade servers in one example. The devices 112 include one or more other components, such as a networking component (e.g., a gateway, a router, a switch), a data storage component (e.g., one or more magnetic disk drives), a cooling component (e.g., a fan), and so forth, in some examples.

The devices 112 are located within the racks 110 of the data center 102. The rack 110 is a physical structure or housing into which multiple chassis can be inserted, mounted, or otherwise placed. The rack 110 includes different physical locations where a chassis of a particular size (referred to as a rack unit or RU) can be placed. Different types of racks 110 can hold different numbers of chassis. In some examples, the rack 110 is configured to hold 50 chassis, 90 chassis, and so forth. The chassis in turn can house a variety of different components, such as the device 112 or the rack power unit 114. Each rack 110 includes one or more data buses, one or more control buses, and one or more power buses that allow data and control information to be communicated to and from the devices 112, and allow power to be communicated to devices 112.

It should be noted that the rack power unit 114 can take different configurations. In one example, the rack power unit 114 is configured in a power shelf configuration (as described in more detail herein in connection with FIGS. 3 and 4) that allows for hot-swap insertion to allow power module exchange under live power operation. However, the

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present disclosure can be implemented in different configurations. For example, the rack power unit **114** can be configured such that AC power is provided to a power distribution unit (PDU) that is then distributed to one or more power supply units (PSUs) in the rack **110**.

In one example, each rack **110** includes one or more of the rack power units **114**. Each rack power unit **114** receives AC power, which can be the external AC power **106** or AC power from the backup generator **104**. Each rack power unit **114** converts the received AC power into DC power, and provides the DC power to the devices **112** within the same rack as that rack power unit. For example, the rack power unit **114(1)** provides DC power to devices **112(1)** in rack **110(1)**, but does not provide DC power to devices **112(n)** in other racks (n). Additionally, although each rack **110** is illustrated in FIG. 1 as including one rack power unit **114**, alternatively a rack **110** can include two or more rack power units **114** that each provide DC power to devices within the same rack as the two or more rack power units.

Although the rack power units **114** are illustrated as receiving AC power, in other examples, the rack power units **114** receive a DC power input. In such examples, rather than (or in addition to) converting received AC power into DC power, the rack power units convert received DC power to a desired voltage (e.g., as desired for a DC power bus within the rack **110**).

Each rack power unit **114** also includes one or more batteries or battery packs and operates as a UPS for devices **112** in the same rack as the rack power unit **114**. In the event of an interruption in AC power received by the rack power unit **114**, the rack power unit **114** draws power from the one or more battery packs to provide to the devices **112**. Thus, if external AC power **106** is interrupted, the rack power unit **114** draws power from the one or more battery packs to provide power to the devices **112** until, for example, the backup generator **104** provides AC power to the rack power unit **114**.

FIG. 2 is a block diagram illustrating an example UPS **200** in accordance with one example. The UPS **200** is an example of or can be embodied as the rack power unit **114** of FIG. 1. The UPS **200** receives AC power **202**, which can be from a variety of sources, such as a power station, a backup generator, and so forth. Although illustrated as AC power, the power **202** can alternatively be DC power (e.g., from a backup generator providing DC power). As discussed herein, situations can arise where there is an interruption in the AC power **202** from one source (e.g., a power station), followed by a time period until the AC power **202** is provided by another source (e.g., a backup generator).

The UPS **200** includes one or more battery backups **204**. The number of battery backups **204** in the UPS **200** can vary. For example, multiple battery backups **204** can be included in the UPS **200** for redundancy (e.g., in the event of a failure of one of the battery backups **204**). In the illustrated example, a single battery backups **204** is shown and is a lithium-ion battery. However, different types of batteries can be used, such as sealed lead-acid batteries.

The UPS **200** includes an AC/DC converter, which is configured as a rectifier/charger **206** in the illustrated example. In operation, the rectifier/charger **206** converts AC power, which periodically changes direction, to DC power, which flows in only one direction. The output of the rectifier/charger **206** is connected to the battery backup **204** and converts the AC power **202** to DC power to charge the battery backup **204** in some examples. The output of the battery backup **204** is also connected to an inverter **208**, which is configured as a power inverter in some examples.

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The inverter **208** converts the DC power from the rectifier/charger **206** back to AC power, which is then provided to a first output **210**. That is, a normal flow double conversion power path is defined from the AC power **202** to the first output **210**. Thus, the first output **210** supplies AC power, such as to AC input IT equipment **212**. However, as should be appreciated, the first output **210** can provide AC power to any device capable of being powered by AC power.

The UPS **200** also includes a power path to a second output **214** that supplies DC power as an output. More particularly, the output of the battery backup **204** is connected to stepdown converter **216** (also referred to as a buck converter). In the illustrated example, the stepdown converter **216** converts the power supplied from the battery backup **204** to a defined lower level, which in this example is 12V. The stepdown converter **216** is a DC-to-DC power converter that steps down voltage (while stepping up current) from the input to the output thereof, in one example. In operation, this power path provides a stepped down DC power supply to the second output **214**, which can be used to provide DC power to DC devices. In the illustrated example, the second output **214** is connected to a DC distribution bus bar **218**, which provides DC power to various devices in the power rack, such as the rack **110** (shown in FIG. 1). It should be noted that in the illustrated example, the output of the AC input IT equipment **212** is also connected to the DC distribution bar **218**.

Thus, in the illustrated configuration, received AC power **202** is converted to DC power, and the DC power is output from the second output **214**, such that this power is provided to the devices in the same rack. It should be noted that the DC power can be any of a variety of different voltages, such as 12 volts, 24 volts, 48 volts, etc. and is provided to the devices in the same rack via the DC distribution bar **218**. That is, each device in the same rack is coupled to the DC distribution bar **218**. As such, rather than converting received AC power to DC power, each device in the rack receives DC power via the DC distribution bar **218**.

It should be noted that the various components of the UPS **200** can be differently configured, such as based on the particular application, power requirements, etc. For example, the particular configuration of the battery backup **204**, the rectifier/charger **206**, the inverter **208**, and the stepdown converter **216** can be varied to meet the requirements for a particular application.

In operation, the UPS **200** is thereby configured as a dual output power supply, namely having the first output **210** and the second output **214**. The first output **210** of the dual output power supply is configured to provide AC output power that is powered through the rectifier/charger **206** (and to the battery backup **204**) and then through the inverter **208**. Additionally, the second output **214** of the dual output power supply is configured to provide DC output power through the rectifier/charger **206**, to the power supply **204**, and then to the stepdown converter **216**. Thus, dual output UPS power that includes switchable AC output power and DC power provides extra redundancy in providing continuous power, including increased power availability and uptime.

In a normal operating state or condition, the UPS **200** is configured to output the AC output from the first output **210** to, for example the power shelf, while switching off the second output **214**. In the case of a power anomaly situation, the second output **214** is configured to source DC power to the DC distribution bar **218** (configured as a bus bar) that is current shared with the power shelf output in one example, or not current shared and provides power alone in another example.

When input AC power instability occurs (e.g., detected by a power detection device), the power shelf operates in the double conversion mode (AC-to-DC-to-AC conversions) as discussed herein and is ready to source from the battery backup **204**. In a situation wherein the power shelf fails, and power is needed for the AC input IT equipment **212** in the rack, the UPS **200** switches on (activates) the second output **214** and sources 12V DC power from the second output **214** until the power supply **214** is depleted or necessary shut down operations are completed (e.g., a non-volatile dual in-line memory (NVDIMM) module or other random-access memory save completes).

In a situation where the rack, such as the AC input IT equipment **212**, is drawing more power than specified by the power cord for the rack (e.g., exceeding the rated power of the power cord), the UPS **200** enables a peak shaving mode to help with the output load. In this mode of operation, the second output **214** current shares with the 12V output of the power shelf.

The UPS **200** also includes a static switch **220** and a bypass switch **222** in some examples. These switches are configured to select different modes of operation. For example, the static switch **220** (e.g., a semiconductor switch) allows current flow therethrough in a normal power flow (high efficiency) mode of operation, and also prevents backflow of current. In this mode of operation, the AC power **202** is connected to the first output **210**. The bypass switch **222** is switchable to select a bypass mode of operation, such as when performing maintenance operations on the UPS **200**. That is, in the bypass mode, the power conversion components are bypassed.

It should be noted that the battery backup **204** in some examples is a plurality of battery packs that provide DC power to devices in the same rack via the DC distribution bar **218**. In the rack-based UPS **200** discussed herein, the battery packs are placed in the same rack as (and thus in close physical proximity to) the devices being powered by the battery packs. This close physical proximity reduces (relative to environments in which the battery packs are further from the devices being powered) losses that can be incurred as power is transferred to the devices.

Furthermore, because power is distributed within the rack via the DC distribution bar **218**, conversions between AC and DC power need not be performed when providing power from battery packs to the devices in the rack. Rather, the power is provided more efficiently because power losses that can be incurred when converting between AC and DC power are not experienced when providing power from the battery packs to the devices in the rack. No additional conversions between AC and DC power need be performed for the sole benefit of battery packs. That is, the battery packs are situated to receive DC power after the received AC power has been converted.

In one example, a rack-based uninterruptible power source is implemented using a single chassis, with the components included in that single chassis. Alternatively, one or more of the components can be implemented across multiple chassis, such as various components being implemented in one or more chassis. In some examples, the DC distribution bar **218** is configured as a power bus having multiple ports that are coupled to one or more computing devices powered by the UPS **200** (e.g., the DC distribution bar **218** can have multiple receptacles that are physically plugged into by the devices, or can have multiple cords and plugs that are plugged into receptacles of the devices). In one example, a power supply controller (e.g., the PDU) can manage the ports individually, allowing DC power to be

turned on or turned off to a particular device as desired or needed and controlled by the power supply controller. The power supply controller can also monitor the power consumed at the power port, and can use the information obtained from this monitoring in different manners, such as determining an average power usage of the device (and thus of the rack), peak power usage of the device (and thus of the rack), and so forth.

One example of a rack **300** is shown in FIGS. **3** and **4**. The rack **300** includes the UPS **200** and a power shelf (Pshelf) **302**, which are configured to operate as described in more detail herein. That is, the UPS **200** is configured to provide extra redundancy in supplying power, such as to supply 12V power in case of power failure to the rack **300**. As a result, power availability and uptime during a power failure event is extended.

In the illustrated example, the rack **300** is connected to two separate power supply lines **304** and **306** (Feed A and Feed B). In one particular implementation, the power supply line **304** is connected to and provides power from an external power source to the UPS **200** and the power supply line **306** is connected to and provides power from an external power source to the power shelf **302**. As such, the rack **300** is protected from a single point of power failure. It should be noted that the UPS **200** and power shelf **302** also include additional components, such as fans **308** and **310** to cool these devices during operation.

As should be appreciated, the second output **214** (shown in FIG. **2**) is connected to the DC distribution bus bar **218**. It should also be appreciated that the rack **300**, in some examples, includes a detection device that allows for the detection of output (e.g., droop output of the power shelf **302**). This allows for a determination of a power supply failure condition or other power anomaly condition. In some examples, a manual activation element (e.g., button or switch) is provided to allow for switching between the different modes of operation discussed herein.

Thus, the present disclosure provides, for example, the UPS **200** having extra redundancy to critical IT equipment in the data center by using dual outputs. For example, in the event of failure of the power shelf **302**, the UPS **200** activates the second output **214** and sources 12V power therefrom. Additionally, when rack equipment draws more power than a maximum rating, the UPS **200** is configured such that the second output **214** current shares with the 12V output of the power shelf **302**. As such, the dual output UPS **200** provides redundancy in the rack **300** with the power shelf **302** during different anomaly or failure events, such as:

1. During input failure, the power shelf **302** runs on a double conversion mode and source power is provided from battery backup, such as the battery backup **204** in the UPS **200**.
2. During power shelf failure, the UPS **200** activates the second output **214** to power, for example, IT equipment in the rack **300**.
3. When the rack equipment is drawing excess power or more power than a rated level, the second output **214** of the UPS **200** current shares with the 12V power output of the power shelf **302**.

Various examples provide dual output UPS, such as the UPS **200** that is configured for multi-mode operation **500** as illustrated in FIG. **5**. As discussed herein, the UPS **200** provides different sources of power depending on the particular operating condition (e.g., normal operation or anomaly operation) by operating in different modes **502**. It should be appreciated that while FIG. **5** illustrates a plurality

of defined modes **502**, additional, fewer, or different modes of operation are contemplated.

In the illustrated example, four modes of operation are possible:

1. High efficiency mode of operation.
2. Double conversion mode of operation.
3. Second output backup power mode of operation (i.e., power output from the second output **214**).
4. Peak shaving mode of operation.

These modes of operation are discussed in more detail herein and are possible as a result of the dual outputs of the UPS **200**. The modes of operation allow for continuous power to be supplied during power anomaly conditions. As should be appreciated, at some point after the power outage occurs, providing of AC power resumes (whether from the external power source or from a backup generator). It should be noted that the point at which the power anomaly condition occurs (e.g., power failure) or the point at which AC power resumes (and thus the UPS ceases to provide power) can be identified in a variety of manners in the power failure detection technology area.

It also should be appreciated that the output from various examples is useful for managing power supplied to a system **800**, which is described in more detail in relation to FIG. **8**.

FIG. **6** is a flowchart illustrating exemplary operations involved in a method **600** for providing a UPS in accordance with various examples. By providing the UPS, power availability and uptime during a power anomaly condition can be increased. The operations illustrated in the flowchart described herein can be performed in a different order than is shown, can include additional or fewer steps and can be modified as desired or needed. Additionally, one or more operations can be performed simultaneously, concurrently or sequentially.

More particularly, and with reference also to FIGS. **1-5**, the method **600** includes configuring a first output to output an AC power supply from an AC power source at **602**. For example, as described herein, the UPS **200** is configured to output AC power at the first output **210**. As such, during normal power conditions, the first output **210** provides AC power, for example, to AC input IT equipment.

The method **600** also includes configuring a second output to output a DC power supply from the AC power source at **604**. For example, as described herein, the UPS is configured to output DC power at the second output **214**. As such, during power anomaly conditions, the second output **214** provides power or current shares power to, for example, DC devices or components in a rack.

More particularly, at **604**, the first and second outputs are operable in the UPS **200** such that the second output is activated during a power anomaly condition to (i) during a power failure, provide power from a battery backup unit or (ii) during an excess power condition, power share with the power supply unit. Thus, additional redundancy is provided by the method **600**.

FIG. **7** is a flowchart illustrating exemplary operations involved in a method **700** for providing power during power anomaly conditions in accordance with various examples. By providing the power during these conditions, critical devices can, for example, be properly shut down by providing additional power for a time period after a power failure in a rack.

More particularly, and with reference also to FIGS. **1-6**, the method **700** includes identifying a power anomaly condition at **702**. For example, as described herein, the power anomaly conditions can include, but are not limited to, an input AC power instability, a power shelf failure, or

equipment drawing an excess amount of power, among others. The power anomaly condition is detected in some examples by measuring electrical voltage or current, or other electrical characteristics at the input or output of one or more components in the rack, or at the input of the rack.

The method **700** also includes activating a second output of a dual output UPS at **704**. For example, when the power anomaly condition is detected, and in response thereto, a DC output of the UPS (e.g., the second output **214** of the UPS **200**) is activated (e.g., an AC output of the UPS is no longer providing power). For example, the second output is activated and allows 12V DC power to be provided to IT equipment in the rack until the battery backup is depleted or shut down operations are completed by the IT equipment.

The method **700** further includes outputting power from the second output at **706**. For example, the DC output of the UPS outputs DC power to power devices within the rack when a power failure is detected or power shares with a power supply unit of the rack during excess power times.

It should be noted that some examples are configured as an add-on or modification kit to a multi-mode system UPS. That is, an existing UPS is modifiable according to the various examples described herein, including to perform the methods **600** and **700**.

Thus, various examples provide a dual output UPS with additional operating modes and redundancy to facilitate providing power to equipment during power anomaly conditions.

ADDITIONAL EXAMPLES

Some aspects and examples disclosed herein are directed to an uninterruptible power supply (UPS) comprising:

- a rectifier configured to receive alternating current (AC) power;
- a first output connected to the rectifier through an inverter, the first output configured to output an AC power supply; and
- a second output connected to the rectifier through a battery backup and a stepdown converter, the second output configured to output a direct current (DC) power supply in response to a detected power anomaly condition.

Additional aspects and examples disclosed herein are directed to a dual output uninterruptible power supply (UPS) comprising:

- a first output;
- a second output;
- an alternating current (AC)/direct current (DC) converter configured to convert input AC power to a DC power supplied to a battery backup;
- an inverter configured to convert the DC power to AC power to output at the first output; and
- a stepdown converter configured to step down the DC power to output at the second output.

Additional aspects and examples disclosed herein are directed to a method for providing an uninterruptible power supply (UPS), the method comprising:

- configuring a first output of the UPS to output an AC power supply from an AC power source; and
- configuring a second output of the UPS to output a DC power supply from an AC power source, the first and second outputs being selectively operable in the UPS and the second output activated during a power anomaly condition to (i) output power during the power

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anomaly condition that is provided by a power supply unit during normal operation or (ii) power share with a power supply unit.

Alternatively, or in addition to the other examples described herein, examples include any combination of the following:

wherein the power anomaly condition comprises one of a power failure or an excess power condition.

wherein the battery backup is connected between the rectifier and the inverter.

wherein the UPS is coupled within a rack and further comprising a power supply unit coupled within the rack, wherein the first output is connected to AC powered equipment in the rack and the second output is connected to a DC distribution bus bar, and the DC power supply is sourced to the DC distribution bus bar as a current shared power source with an output of the power supply unit during the power anomaly condition. further comprising a first power supply line connected to the UPS and a second power supply line connected to the power supply unit.

wherein the first output is configured to operate in a high efficiency mode or a double conversion mode.

wherein the second output is activated in response to a power anomaly condition.

wherein the second output is in an active state until one or more shut down operations of connected devices having random-access memory are completed.

wherein the first output is configured to operate in modes of operation that are different than modes of operation of the second output.

wherein the first and second output are configured for selectable operation in a plurality of operating modes comprising a high efficiency mode of operation, a double conversion mode of operation, a second output backup power mode of operation, and a peak shaving mode of operation.

configuring the first and second outputs to selectively operate in a plurality of operating modes comprising a high efficiency mode of operation, a double conversion mode of operation, a second output backup power mode of operation, and a peak shaving mode of operation.

configuring an alternating current (AC)/direct current (DC) converter to convert the AC power source to a DC power supplied to a battery backup.

configuring a stepdown converter to step down a voltage output of the battery backup.

configuring the first and second outputs for operation at a same time.

Example Operating Environment

FIG. 8 is a block diagram of an example system 800 implementing aspects disclosed herein, and is designated generally as a system 800. The system 800 is but one example of a suitable computing environment and is not intended to suggest any limitation as to the scope of use or functionality of the examples disclosed herein. Neither should the system 800 be interpreted as having any dependency or requirement relating to any one or combination of components/modules illustrated. The examples disclosed herein may be described in the general context of computer code or machine-useable instructions, including computer-executable instructions such as program components, being executed by a computer or other machine, such as a personal data assistant or other handheld device. Generally, program components including routines, programs, objects, components, data structures, and the like, refer to code that

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performs particular tasks, or implement particular abstract data types. The disclosed examples may be practiced in a variety of system configurations, including personal computers, laptops, smart phones, mobile tablets, hand-held devices, consumer electronics, specialty computing devices, etc. The disclosed examples may also be practiced in distributed computing environments when tasks are performed by remote-processing devices that are linked through a communications network.

The system 800 includes a bus 810 that directly or indirectly couples the following devices: a computer-storage memory 812, one or more processors 814, one or more presentation components 816, input/output (I/O) ports 818, I/O components 820, a power supply 822 (that includes the UPS 200), and a network component 824. While the system 800 is depicted as a seemingly single device, multiple systems 800 can work together and share the depicted device resources. For instance, the computer-storage memory 812 can be distributed across multiple devices, processor(s) 814 may provide housed on different devices, and so on.

The bus 810 represents one or more busses (such as an address bus, data bus, or a combination thereof). Although the various blocks of FIG. 8 are shown with lines for the sake of clarity, in reality, delineating various components is not so clear, and metaphorically, the lines would more accurately be grey and fuzzy. For example, a presentation component such as a display device can be considered to be an I/O component. Also, processors have memory and FIG. 8 is merely illustrative of an exemplary computing device that can be used in connection with one or more disclosed examples. Distinction is not made between such categories as “workstation,” “server,” “laptop,” “hand-held device,” etc., as all are contemplated within the scope of FIG. 8 and the references herein to a “system”. The computer-storage memory 812 may take the form of the computer-storage media references below and operatively provide storage of computer-readable instructions, data structures, program modules and other data for the system 800. For example, the computer-storage memory 812 can store an operating system, a universal application platform, or other program modules and program data. The computer-storage memory 812 can be used to store and access instructions configured to carry out the various operations disclosed herein.

As mentioned below, the computer-storage memory 812 may include computer-storage media in the form of volatile and/or nonvolatile memory, removable or non-removable memory, data disks in virtual environments, or a combination thereof. And the computer-storage memory 812 may include any quantity of memory associated with or accessible by the computing device 800. The memory 812 may be internal to the computing device 800 (as shown in FIG. 8), external to the computing device 800 (not shown), or both (not shown). Examples of the memory 812 include, without limitation, random access memory (RAM); read only memory (ROM); electronically erasable programmable read only memory (EEPROM); flash memory or other memory technologies; CD-ROM, digital versatile disks (DVDs) or other optical or holographic media; magnetic cassettes, magnetic tape, magnetic disk storage or other magnetic storage devices; memory wired into an analog computing device; or any other medium for encoding desired information and for access by the computing device 800. Additionally, or alternatively, the computer-storage memory 812 may be distributed across multiple computing devices 800, e.g., in a virtualized environment in which instruction processing is carried out on multiple devices 800. For the purposes of this disclosure, “computer storage

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media,” “computer-storage memory,” “memory,” and “memory devices” are synonymous terms for the computer-storage memory **812**, and none of these terms include carrier waves or propagating signaling.

Therefore, for example, a computer storage medium should not be interpreted to be a propagating signal per se. Propagated signals per se are not examples of computer storage media. Although the computer storage medium (the memory **812**) is shown within the system **800**, it will be appreciated by a person skilled in the art, that the storage can be distributed or located remotely and accessed via a network or other communication link (e.g. using a communication interface).

The processor(s) **814** may include any quantity of processing units that read data from various entities, such as the memory **812** or I/O components **820**. Specifically, the processor(s) **814** are programmed to execute computer-executable instructions for implementing aspects of the disclosure. The instructions may be performed by the processor, by multiple processors within the computing device **800**, or by a processor external to the client computing device **800**. In some examples, the processor(s) **814** are programmed to execute instructions such as those illustrated in the flow charts discussed below and depicted in the accompanying drawings. Moreover, in some examples, the processor(s) **814** represent an implementation of analog techniques to perform the operations described herein. For example, the operations may be performed by an analog client computing device **800** and/or a digital client computing device **800**. Presentation component(s) **816** present data indications to a user or other device. Exemplary presentation components include a display device, speaker, printing component, vibrating component, etc. One skilled in the art will understand and appreciate that computer data may be presented in a number of ways, such as visually in a graphical user interface (GUI), audibly through speakers, wirelessly between computing devices **800**, across a wired connection, or in other ways. Ports **818** allow computing device **800** to be logically coupled to other devices including I/O components **820**, some of which may be built in. Examples I/O components **820** include, for example but without limitation, a microphone, joystick, game pad, satellite dish, scanner, printer, wireless device, etc.

The system **800** may operate in a networked environment via the network component **824** using logical connections to one or more remote computers. In some examples, the network component **824** includes a network interface card and/or computer-executable instructions (e.g., a driver) for operating the network interface card. Communication between the computing device **800** and other devices may occur using any protocol or mechanism over any wired or wireless connection. In some examples, the network component **824** is operable to communicate data over public, private, or hybrid (public and private) using a transfer protocol, between devices wirelessly using short range communication technologies (e.g., near-field communication (NFC), Bluetooth™ branded communications, or the like), or a combination thereof. For example, network component **824** communicates over a communication link **826** with a network **828**.

Although described in connection with an example system **800**, examples of the disclosure are capable of implementation with numerous other general-purpose or special-purpose computing system environments, configurations, or devices. Examples of well-known computing systems, environments, and/or configurations that may be suitable for use with aspects of the disclosure include, but are not limited to,

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smart phones, mobile tablets, mobile computing devices, personal computers, server computers, hand-held or laptop devices, multiprocessor systems, gaming consoles, micro-processor-based systems, set top boxes, programmable consumer electronics, mobile telephones, mobile computing and/or communication devices in wearable or accessory form factors (e.g., watches, glasses, headsets, or earphones), network PCs, minicomputers, mainframe computers, distributed computing environments that include any of the above systems or devices, VR devices, holographic device, and the like. Such systems or devices may accept input from the user in any way, including from input devices such as a keyboard or pointing device, via gesture input, proximity input (such as by hovering), and/or via voice input.

Examples of the disclosure may be described in the general context of computer-executable instructions, such as program modules, executed by one or more computers or other devices in software, firmware, hardware, or a combination thereof. The computer-executable instructions may be organized into one or more computer-executable components or modules. Generally, program modules include, but are not limited to, routines, programs, objects, components, and data structures that perform particular tasks or implement particular abstract data types. Aspects of the disclosure may be implemented with any number and organization of such components or modules. For example, aspects of the disclosure are not limited to the specific computer-executable instructions or the specific components or modules illustrated in the figures and described herein. Other examples of the disclosure may include different computer-executable instructions or components having more or less functionality than illustrated and described herein. In examples involving a general-purpose computer, aspects of the disclosure transform the general-purpose computer into a special-purpose computing device when configured to execute the instructions described herein.

By way of example and not limitation, computer readable media comprise computer storage media and communication media. Computer storage media include volatile and nonvolatile, removable and non-removable memory implemented in any method or technology for storage of information such as computer readable instructions, data structures, program modules, or the like. Computer storage media are tangible and mutually exclusive to communication media. Computer storage media are implemented in hardware and exclude carrier waves and propagated signals. Computer storage media for purposes of this disclosure are not signals per se. Exemplary computer storage media include hard disks, flash drives, solid-state memory, phase change random-access memory (PRAM), static random-access memory (SRAM), dynamic random-access memory (DRAM), other types of random-access memory (RAM), read-only memory (ROM), electrically erasable programmable read-only memory (EEPROM), flash memory or other memory technology, compact disk read-only memory (CD-ROM), digital versatile disks (DVD) or other optical storage, magnetic cassettes, magnetic tape, magnetic disk storage or other magnetic storage devices, or any other non-transmission medium that can be used to store information for access by a computing device. In contrast, communication media typically embody computer readable instructions, data structures, program modules, or the like in a modulated data signal such as a carrier wave or other transport mechanism and include any information delivery media.

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Any range or device value given herein can be extended or altered without losing the effect sought, as will be apparent to the skilled person.

Although the subject matter has been described in language specific to structural features and/or methodological acts, it is to be understood that the subject matter defined in the appended claims is not necessarily limited to the specific features or acts described above. Rather, the specific features and acts described above are disclosed as example forms of implementing the claims.

It will be understood that the benefits and advantages described above can relate to one embodiment or can relate to several embodiments. The embodiments are not limited to those that solve any or all of the stated problems or those that have any or all of the stated benefits and advantages. It will further be understood that reference to 'an' item refers to one or more of those items.

The examples illustrated and described herein as well as examples not specifically described herein but within the scope of aspects of the claims constitute exemplary means for providing a UPS.

The term "comprising" is used in this specification to mean including the feature(s) or act(s) followed thereafter, without excluding the presence of one or more additional features or acts.

In some examples, the operations illustrated in the figures can be implemented as software instructions encoded on a computer readable medium, in hardware programmed or designed to perform the operations, or both. For example, aspects of the disclosure can be implemented as a system on a chip or other circuitry including a plurality of interconnected, electrically conductive elements.

The order of execution or performance of the operations in examples of the disclosure illustrated and described herein is not essential, unless otherwise specified. That is, the operations can be performed in any order, unless otherwise specified, and examples of the disclosure can include additional or fewer operations than those disclosed herein. For example, it is contemplated that executing or performing a particular operation before, contemporaneously with, or after another operation is within the scope of aspects of the disclosure.

When introducing elements of aspects of the disclosure or the examples thereof, the articles "a," "an," "the," and "said" are intended to mean that there are one or more of the elements. The terms "comprising," "including," and "having" are intended to be inclusive and mean that there can be additional elements other than the listed elements. The term "exemplary" is intended to mean "an example of." The phrase "one or more of the following: A, B, and C" means "at least one of A and/or at least one of B and/or at least one of C."

Having described aspects of the disclosure in detail, it will be apparent that modifications and variations are possible without departing from the scope of aspects of the disclosure as defined in the appended claims. As various changes could be made in the above constructions, products, and methods without departing from the scope of aspects of the disclosure, it is intended that all matter contained in the above description and shown in the accompanying drawings shall be interpreted as illustrative and not in a limiting sense.

What is claimed is:

1. A rack comprising:

a power shelf configured to receive power from a first power source or a second power source, and provide alternating current (AC) power and direct current (DC) power to a device in the rack;

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the first power source configured to provide AC power to the power shelf; and

the second power source comprises an uninterruptable power supply (UPS) that is configured to, upon failure of the first power source, receive input DC power from a battery, convert the input DC power to an AC power, and provide the AC power to the power shelf, the UPS being further configured to:

current share DC power with an output of the power shelf upon the device drawing more DC power than a rated level from the output of the power shelf, and provide DC power to the device upon failure of the rack.

2. The rack of claim 1, wherein a step down converter is used to step down DC voltage of the battery to a lower DC voltage to provide DC power to the device upon failure of the rack.

3. The rack of claim 1, wherein the battery is charged by a rectifier connected directly to the first power source.

4. The rack of claim 1, wherein the first power source comprises a power station managed by a power utility company or a power generator run on local fuel.

5. The rack of claim 1 further comprising a first power supply line connected to the UPS and a second power supply line connected to the power shelf.

6. The rack of claim 1, wherein the failure of the first power source comprises an instability in the first power source.

7. The rack of claim 1, wherein the UPS continues to provide DC power to the device upon failure of the rack at least until necessary shut down operations of the device is completed.

8. The rack of claim 1, wherein during an instability in the first power source, the power shelf operates in a double conversion mode.

9. The rack of claim 1, wherein the first power source is connected to the power shelf through a static switch that allows current flow therethrough in a normal power flow mode of operation, and prevents backflow of current.

10. The rack of claim 1, wherein the rack further includes a bypass switch that, when turned on, bypasses power conversion components of the UPS, allowing maintenance of the UPS.

11. A method for providing a rack, the method comprising:

configuring a power shelf of the rack to receive power from a first power source or a second power source, and provide alternating current (AC) power and direct current (DC) power to a device in the rack;

configuring the first power source to provide AC power to the power shelf;

configuring the second power source as an uninterruptable power supply (UPS) that, upon failure of the first power source, receives input DC power from a battery, converts the input DC power to an AC power, and provides the AC power to the power shelf; and

configuring the UPS to further:

current share DC power with an output of the power shelf upon the device drawing more DC power than a rated level of a power cord for the power shelf, and provide DC power to the device upon failure of the rack.

12. The method of claim 11, wherein a step down converter is used to step down DC voltage of the battery to a lower DC voltage to provide DC power to the device upon failure of the rack.

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13. The method of claim 11, wherein the battery is charged by a rectifier connected directly to the first power source.

14. The method of claim 11, wherein the first power source comprises a power station managed by a power utility company or a power generator run on local fuel. 5

15. The method of claim 11 further comprising configuring a first power supply line to be connected to the UPS and a second power supply line to be connected to the power shelf.

16. The method of claim 11, wherein the failure of the first power source comprises an instability in the first power source. 10

17. The method of claim 11, wherein the UPS continues to provide DC power to the device upon failure of the rack at least until necessary shut down operations of the device are completed. 15

18. The method of claim 11, wherein during an instability in the first power source, the power shelf operates in a double conversion mode.

19. The method of claim 11, wherein the first power source is connected to the power shelf through a static 20

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switch that allows current flow therethrough in a normal power flow mode of operation, and prevents backflow of current.

20. A rack comprising:

a power shelf configured to receive power from a first power source or a second power source, and provide alternating current (AC) power and direct current (DC) power to a device in the rack;

the first power source configured to provide AC power to the power shelf, and

the second power source comprises an uninterruptable power supply (UPS) that is configured to, upon failure of the first power source, receive input DC power from a battery, convert the input DC power to an AC power, and provide the AC power to the power shelf,

the UPS and the power shelf being further configured to both provide power to the device upon the device drawing more DC power than a rated level of a power cord for the power shelf.

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